Lab 01

Python for engineers

03/30/2026

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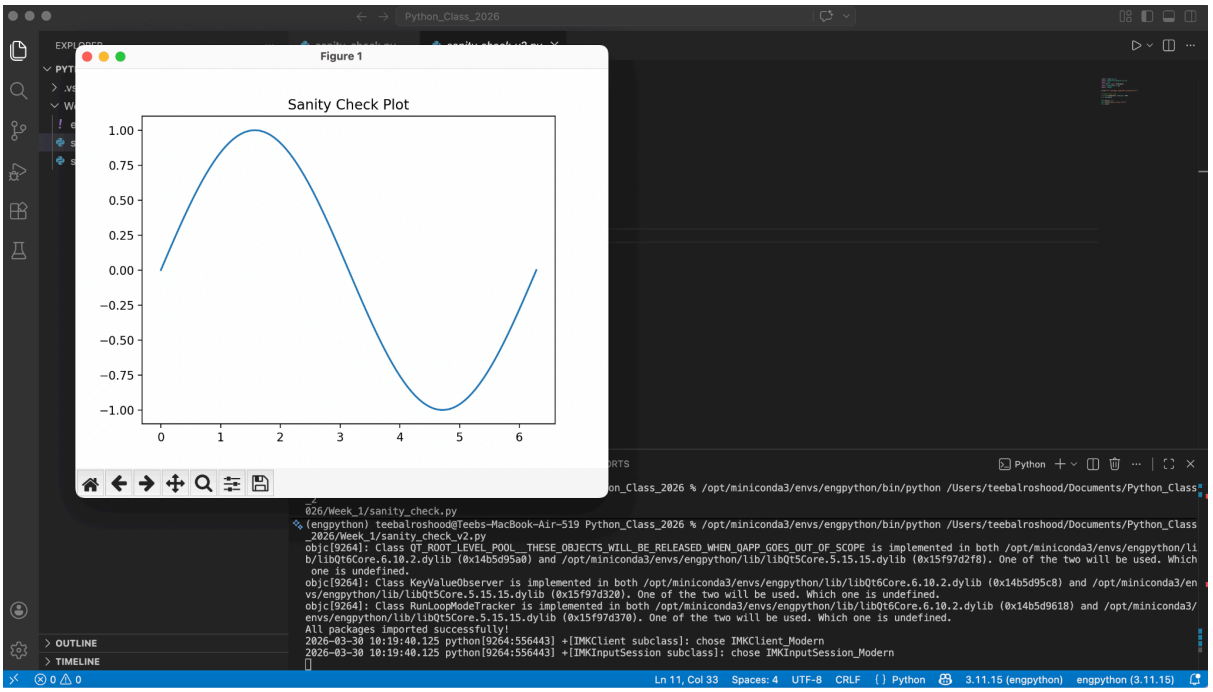
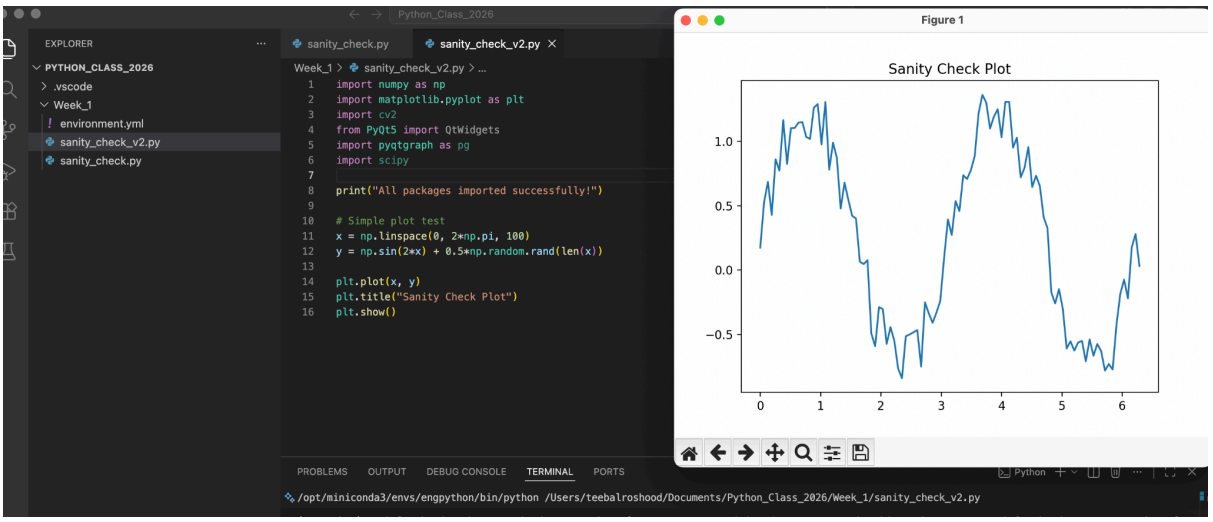
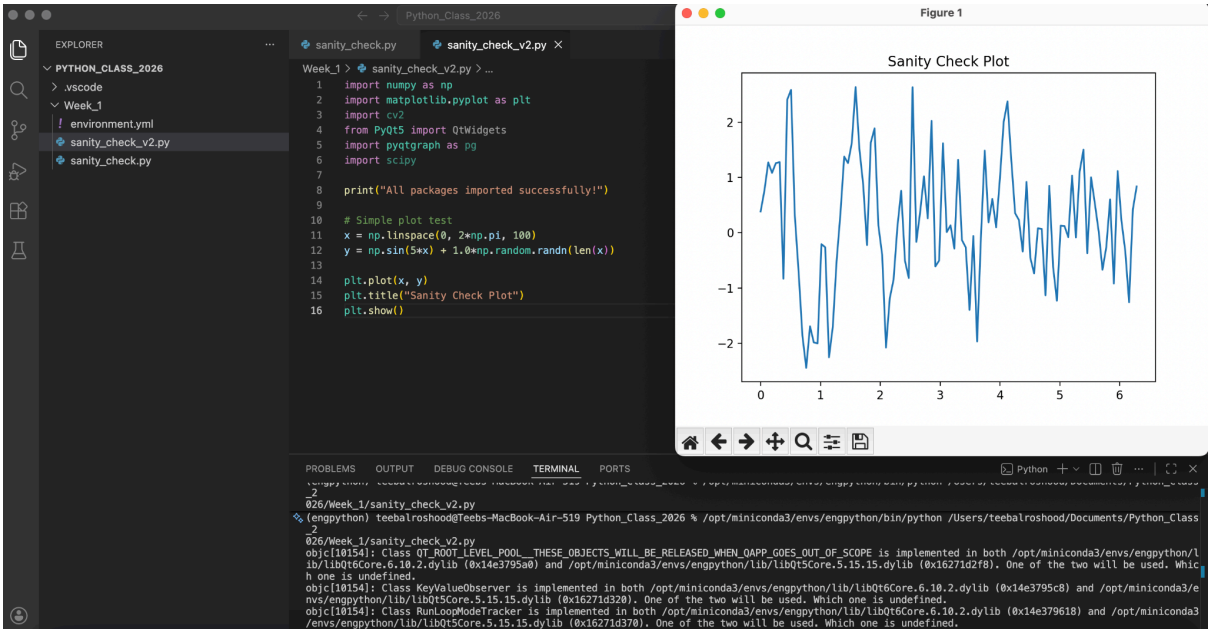
Part 1:



Part 2:



Changed frequency and noise level:



sanity_check.py sanity_check_v2.py

Week_1 > sanity_check_v2.py > ...

~/Documents/Python_Class_2026/Week_1/sanit

```python
import nump
import matplotlib.pyplot as plt
import cv2
from PyQt5 import QtWidgets
import pyqtgraph as pg
import scipy

print("All packages imported successfully!")

# Simple plot test
x = np.linspace(0, 2*np.pi, 100)
y = np.sin(5*x) + 1.0*np.random.randn(len(x))

plt.plot(x, y)
plt.title("Sanity Check Plot")
plt.show()
```

Modified code

Reflection:

- **What effect does noise have on the signal?**
  Noise adds random variations to the signal, making it less smooth and more irregular. It can distort the original shape of the sine wave and hide important patterns. As noise increases, it becomes harder to clearly identify the true signal.
- **How does increasing frequency change the plot?**
  Increasing the frequency makes the signal oscillate faster, resulting in more waves within the same interval. The plot appears

more compressed horizontally because the cycles occur more frequently. This makes the signal change more rapidly over time.

- **What would make this data difficult to analyze?**
  High noise levels can obscure the original signal, making patterns difficult to detect. Very high frequency can also make the signal appear cluttered and harder to interpret. A combination of both high noise and high frequency makes analysis especially challenging.